```
Experiment 4
SQL*Plus: Release 11.2.0.1.0 Production on Tue Sep 8 14:29:09 2026

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Enter user-name: A044@db2021
Enter password:

Connected to:
Oracle Database 11g Release 11.2.0.1.0 - Production

SQL> create table student21(
  2 student_id number(5) primary key,
  3 student_name varchar2(30),
  4 department varchar2(20),
  5 marks number(3)
  6 );

Table created.

SQL> insert into student21 values('&student_id','&student_name','&department','&marks');
Enter value for student_id: 1
Enter value for student_name: diya
Enter value for department: cse
Enter value for marks: 90
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('1','diya','cse','90')

1 row created.

SQL> /
Enter value for student_id: 2
Enter value for student_name: omar
Enter value for department: ece
Enter value for marks: 95
old    1: insert into student21 values('&student_id','&student_name','&department','&marks')
new    1: insert into student21 values('2','omar','ece','95')

1 row created.

SQL> /
Enter value for student_id: 3
Enter value for student_name: sita
Enter value for department: civil
Enter value for marks: 75
old    1: insert into student21 values('&student_id','&student_name','&department','&marks')
new    1: insert into student21 values('3','sita','civil','75')

1 row created.

SQL> /
Enter value for student_id: 4
Enter value for student_name: neha
Enter value for department: IT
Enter value for marks: 85
old    1: insert into student21 values('&student_id','&student_name','&department','&marks')
new    1: insert into student21 values('4','neha','IT','85')

1 row created.

SQL> /
Enter value for student_id: 5
Enter value for student_name: ravi
Enter value for department: mech
Enter value for marks: 92
old    1: insert into student21 values('&student_id','&student_name','&department','&marks')
new    1: insert into student21 values('5','ravi','mech','92')

1 row created.

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 diya                           cse                          90
         2 omar                           ece                          95
```

```
         3 sita                           civil                     75
         4 neha                           IT                        85
         5 ravi                           mech                      92

SQL> select count(*) as total_students from student;
select count(*) as total_students from student
                                       *
ERROR at line 1:
ORA-00942: table or view does not exist


SQL> select count(*) as total_students from student21;

TOTAL_STUDENTS
--------------
             5

SQL> select sum(marks) as total_marks from student21;

TOTAL_MARKS
-----------
        437

SQL> select avg(marks) as average_marks from student21;

AVERAGE_MARKS
-------------
         87.4

SQL> select avg(marks) as total_marks from student21;

TOTAL_MARKS
-----------
       87.4

SQL> select min(marks) as total_marks from student21;

TOTAL_MARKS
-----------
         75

SQL> select min(marks) as min_marks from   student21;

 MIN_MARKS
----------
        75

SQL> select max(marks) as total_marks from student21;p
  2 select max(marks) as total_marks from student21;
select max(marks) as total_marks from student21;p
                                                 *
ERROR at line 1:
ORA-00911: invalid character


SQL> select max(marks) as total_marks from student21;

TOTAL_MARKS
-----------
         95

SQL> select max(marks) as highest
  2
SQL>
SQL> select max(marks) as highest_marks from student21;

HIGHEST_MARKS
-------------
           95

SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
```

```
ece                          95
mech                         92

SQL>
SQL> /

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
ece                          95
mech                         92

SQL> insert into student21 values('&student_id','&student_name','&department','&marks');
Enter value for student_id: 6
Enter value for student_name: uma
Enter value for department: cse
Enter value for marks: 80
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('6','uma','cse','80')

1 row created.

SQL> /
Enter value for student_id: 7
Enter value for student_name: kavya
Enter value for department: ece
Enter value for marks: 97
old    1: insert into student21 values('&student_id','&student_name','&department','&marks')
new    1: insert into student21 values('7','kavya','ece','97')

1 row created.

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 diya                           cse                          90
         2 omar                           ece                          95
         3 sita                           civil                        75
         4 neha                           IT                           85
         5 ravi                           mech                         92
         6 uma                            cse                          80
         7 kavya                          ece                          97

7 rows selected.

SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          85
civil                        75
ece                          96
mech                         92

SQL> select department,avg(marks) as avg_marks from student21;
select department,avg(marks) as avg_marks from student21
       *
ERROR at line 1:
ORA-00937: not a single-group group function


SQL> select department,avg(marks0 as total_words from student21;
select department,avg(marks0 as total_words from student21
                              *
ERROR at line 1:
ORA-00907: missing right parenthesis


SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
```

```
cse                          85
civil                        75
ece                          96
mech                         92

SQL> select department,avg(marks) as total_marks from student21 group by department;

DEPARTMENT           TOTAL_MARKS
-------------------- -----------
IT                            85
cse                           85
civil                         75
ece                           96
mech                          92

SQL> select department,count(marks) as count_student from student21 group by department;

DEPARTMENT           COUNT_STUDENT
-------------------- -------------
IT                               1
cse                              2
civil                            1
ece                              2
mech                             1

SQL> select department,max(marks) as max_marks from student21 group by department;

DEPARTMENT            MAX_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
ece                          97
mech                         92

SQL> select department,avg(marks) as avg_marks from student21 group by department having avg(marks) > 85;

DEPARTMENT            AVG_MARKS
-------------------- ----------
ece                          96
mech                         92

SQL> select department,count(*) as total_students from student21 group by department having count(*)>1;

DEPARTMENT           TOTAL_STUDENTS
-------------------- --------------
cse                               2
ece                               2

SQL> select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks) from st
    udent21 group by department having avg(marks)>80;
select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks) from student
    21 group by department having avg(marks)>80
                                                                                                *
ERROR at line 1:
ORA-00923: FROM keyword not found where expected


SQL> select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks from stu
    dent21 group by department having avg(marks)>80;

DEPARTMENT           NO_OF_STUDENTS AVG_MARKS MAX_MARKS
-------------------- -------------- ---------- ----------
IT                                1         85         85
cse                               2         85         90
ece                               2         96         97
mech                              1         92         92

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 diya                           cse                          90
         2 omar                           ece                          95
         3 sita                           civil                        75
         4 neha                           IT                           85
         5 ravi                           mech                         92
```

```
         6 uma                             cse             80
         7 kavya                           ece             97

7 rows selected.

SQL> create table employee10{
  2 emp_id number(5) primary key,
  3 emp_name varchar2(30),
  4 salary number(8)
  5 );
create table employee10{
                       *
ERROR at line 1:
ORA-00922: missing or invalid option


SQL> create table employee10(
  2 emp_id number(5) primary key,
  3 emp_name varchar2(30),
  4 salary number(8)
  5 );

Table created.

SQL> insert into employee10 values(101,'arun',100000);

1 row created.

SQL> insert into employee10 values(102,'manu',50000);

1 row created.

SQL> grant select on employee10 to scott;

Grant succeeded.

SQL> grant select,insert,update on employee to scott;

Grant succeeded.

SQL> revoke insert on employee from scott;

Revoke succeeded.

SQL> insert into employee10 values(103,'vikram',200000);

1 row created.

SQL> commit;

Commit complete.

SQL> insert into employee10 values(104,'suresh',105000);

1 row created.

SQL> savepoint sp1;

Savepoint created.

SQL> insert into employee10 values(105,'mahesh',100000);

1 row created.

SQL> rollback to sp1;

Rollback complete.

SQL> select * from employee10;

    EMP_ID EMP_NAME                         SALARY
---------- ------------------------------ ----------
       101 arun                           100000
       102 manu                            50000
       103 vikram                         200000
       104 suresh                         105000
SQL> grant select on employee10 to user_name;
```

```
grant select on employee10 to user_name
                              *
ERROR at line 1:
ORA-01917: user or role 'USER_NAME' does not exist


SQL> create table alumni(
  2 student_name varchar(30)
  3 );

Table created.

SQL> insert into alumni values('diya');

1 row created.

SQL> insert into alumni values('omar');

1 row created.

SQL> insert into alumni values('sita');

1 row created.

SQL> select * from student21 where department in ('cse','IT');

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 diya                           cse                          90
         4 neha                           IT                           85
         6 uma                            cse                          80

SQL> select * from student21 where student_id in(1,2);

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 diya                           cse                          90
         2 omar                           ece                          95

SQL> select * from student21 where marks > any (
  2 select marks from student where department = 'ece' );
select marks from student where department = 'ece' )
                 *
ERROR at line 2:
ORA-00942: table or view does not exist


SQL> select * from student21 where marks > any (
  2 select marks from student21 where department = 'ece' );

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         7 kavya                          ece                          97

SQL> select * from student21 where marks > any (select marks from student21 where department='cse');

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         7 kavya                          ece                          97
         2 omar                           ece                          95
         5 ravi                           mech                         92
         1 diya                           cse                          90
         4 neha                           IT                           85
SQL>
```